

MCQs

1. If the per-unit conversion cost was \$1.60 last month and direct materials cost was \$120,000, direct labor costs was \$40,000 and manufacturing overhead costs was \$96,000. How many units did they manufacture during the month?

- A) 100,000
- B) 25,000
- C) 85,000
- D) 160,000
- E) 60,000

Answer: [_____]

$$(\$40,000 + \$96,000) / X = \$1.60$$

$$X = 85,000$$

2. Trendy Wear Company has exclusive rights to sell the Blueberry Shortcake T-shirts. The company currently pays sales commission of 5% on each shirt sold. If a graph is prepared with the vertical axis representing per-unit cost and the horizontal axis representing units sold, how would a line that depicts sales commissions be drawn?

- A) As a straight diagonal line, sloping upward to the right.
- B) As a straight diagonal line, sloping downward to the right.
- C) As a vertical line.
- D) As a horizontal line.
- E) As a curvilinear line.

Answer: [_____]

3. Cold Company has the following data for Product X:

BE in units	375
Sales in units	1,000
Total sales dollar	\$750,000
Net Operating Income (NOI)	300,000

The fixed costs for Product X is

- A) \$150,000
- B) \$180,000
- C) \$200,000
- D) \$450,000
- E) None of the above

$$\text{Sales} - \text{VC} - \text{FC} = \text{Profit}$$

$$\text{CM} - \text{FC} = \text{Profit}$$

$$\text{CM} = \text{Profit} + \text{FC}$$

$$\text{BE ratio} = \frac{\text{FC}}{\text{CM}} = \frac{\text{FC}}{\text{FC} + \text{Profit}}$$

$$\frac{375}{1,000} = \frac{\text{FC}}{\text{FC} + 300,000}$$

$$625 \text{ FC} = 375 (300,000)$$

$$\text{FC} = \$180,000$$

4. Ray Corporation recently computed total product costs of \$888,000 and total period costs of \$333,000, excluding \$33,000 of sales commissions that were overlooked by the company's administrative assistant. On the basis of this information, the correct operating expenses in the income statement should be:

- A) \$ 33,000
- B) \$333,000
- C) \$366,000
- D) \$888,000
- E) \$921,000

Answer: [_____]

As the question asks for the correct operating expenses (same as period costs or selling and admin), we adjust for the omission of sales commission of \$33,000. Hence, the correct operating expenses = \$333k + \$33k = \$366k.

5. Which of the following costs is not a component of manufacturing overhead?

- A) Factory utilities
- B) Cost of factory equipment
(Cost of the equipment itself is not (a capital item) but depreciation is, need to review all choices and pick the best answer.)
- C) Indirect labor
- D) Indirect materials
- E) Property taxes on the manufacturing plant

Answer: [_____]

6. Which of the followings will **not** increase a company's breakeven point?

- A) Increasing its total fixed costs
- B) Decreasing contribution margin per unit
- C) Increasing variable cost per unit
- D) Increasing the selling price per unit
- E) None of the above

Answer: [_____]

7. When closing overapplied manufacturing overhead to cost of goods sold, which of the following would be true?

- A) Work in process will decrease.
- B) Cost of goods sold will increase.
- C) Net income will decrease.
- D) Finished goods will decrease.
- E) Gross margin will increase.

Answer: [_____]

8. Which of the following position of a bank is expected to provide and use management accounting information?

- A. Financial controller.
- B. Marketing director.
- C. Head of human resource.
- D. Head of information technology manager.
- E. All of the above.

Answer: [_____]

9. Which of the following statements about prime costs and conversion costs is **true**?

- A. The difference between prime cost and conversion cost can be calculated by finding the difference between the cost of direct materials and manufacturing overheads.
- B. Prime costs and conversion costs are product costs.
- C. Period costs can be calculated by subtracting product costs from total costs.
- D. Both "B" and "C" are true.
- E. All of the above are true.

Answer: [_____]

10. Which of the following statement is true:

- A. Break-even for a multiple-product firm in dollars can be calculated by dividing total fixed costs by the contribution margin of a composite unit. (Should be "by the CM ratio")
- B. Break-even for a multiple-product firm in dollars can be calculated by dividing manufacturing fixed costs by the contribution margin ratio of a composite unit. (Should be "total fixed costs" and not "manufacturing fixed costs")
- C. Break-even for a multiple-product firm can only be calculated when the percentage of products sold is the same for all products.
- D. Break-even for a multiple-product firm can be calculated by multiplying total fixed costs by the contribution margin ratio of the product with the highest percentage in the sales mix.
- E. None of the above.

Answer: [_____]

11. What is operating leverage?

- A. A measure of how a change in sales volume will affect profits.
- B. A measure of how a change in debt will affect profits.
- C. A measure of how a change in sales price will affect profits.
- D. Both "B" and "C" above.
- E. All of the above.

Answer: [_____]

12. Which of the following statement(s) is/(are) true regarding overhead allocation?

- A. Actual overheads may not be known until days or weeks after the end of the period, therefore using predetermined overhead rates avoids delaying the calculation of unit cost for reporting purposes.
- B. Many types of overhead costs are fixed even though output fluctuates during the period; using predetermined overhead rates reduce fluctuations in overhead rate that result from changes in production quantity.
- C. Using predetermined overhead rates allows the cost to be estimated in a timely manner for decision making purposes.
- D. Only "B" and "C"
- E. "A" and "B" and "C"

Answer: [_____]

13. Which of the following statements about process costing is **false**?

- A. The individual units of production manufactured in a process costing environment are likely to be homogeneous and indistinguishable from one another.
- B. Average cost per equivalent unit is computed for each processing department.
- C. The weighted-average inventory method isolates previous period costs from those costs incurred in the current period.
- D. When there is no beginning and ending work-in-process, total equivalent units of production is the same as the total physical units of production for the period.
- E. Transferred-in materials are treated as 100% added in the current department.

Answer: [_____]

14. Public Relations in a manufacturing company is an example of a/(an):

- A. unit-level activity.
- B. batch-level activity.
- C. product-level activity.
- D. customer-level activity.
- E. organization-sustaining activity.

Answer: [_____]

15. Which of the following statements is **true**?

- A. Direct costs are always variable and never fixed in nature.
- B. All costs traceable to the cost object must always be recorded as direct costs.
- C. The classification of direct and indirect costs is used in manufacturing, non-manufacturing and not-for-profit organisations.
- D. The concept of direct and indirect costs applies only to product costs and not period costs.
- E. Fixed costs are always indirect costs regardless of the cost object.

Answer: [_____]

16. BB and Baba manufactures school uniforms. It uses a job costing system by tracing all direct manufacturing costs to each job and allocates indirect manufacturing costs to each job based on a predetermined manufacturing overhead rate. The Company has the following estimated costs for next year:

Direct materials	\$15,000
Direct labour	\$55,000
Sales commissions	\$7,000
Salary of production supervisor	\$35,000
Indirect materials	\$5,000
Advertising expense	\$1,000
Depreciation on factory equipment	\$13,000
Factory rent	\$16,000
Other manufacturing overheads	\$3,000

The company estimates that 10,000 direct labour hours will be worked during the year. If manufacturing overhead is applied on the basis of direct labour-hours, the manufacturing overhead rate per hour will be:

- A. \$3.20
- B. \$6.70
- C. \$7.20**
- D. \$8.00
- E. \$15.00

Answer: [_____]

Estimated manufacturing overhead

Salary of production supervisor	\$35,000
Indirect materials	\$5,000
Depreciation on factory equipment	\$13,000
Factory rent	\$16,000
Other manufacturing overheads	\$3,000
Total	\$72,000

Predetermined manufacturing overhead rate

=

$$\text{\$72,000} \div 10,000\text{DLH}$$

$$= \text{\$7.20 per DLH}$$

Structured Questions

1. Harry estimates his monthly total cost (Y) of using the Bloomberg database as: $Y = \$3.00X + \50 , where X is the number of searches on Bloomberg made by Harry. Harry is expecting to do 90 searches in June, what will be the expected total cost for June regarding the use of the Bloomberg database?

Answer: [_____]

ANSWER

$$Y = \$3 * 90 + \$50 = \$320$$

2. Bean & Co provides accountancy services, earning \$100 per hour worked. It hires accountants on contracts that require three months' notice for termination of employment. The accountants are paid a total basic salary of \$30,000 per month and a bonus of 10% of total revenue. Bean & Co operates from rented premises and the lease is due to expire on 31 December 2012.

It is now late September 2012 and the managing partner of Bean & Co is budgeting for the months of October to December 2012:

<u>Budget for Oct – Dec 2012</u>	\$
Revenue (\$100 x 3,000 hours)	300,000
Basic salaries	(90,000)
Employee bonus (10% of revenue)	(30,000)
Rental	(120,000)
Profit	<u>60,000</u>

What is the total contribution margin?

Answer: [_____]

ANSWER

$$\text{Contribution margin} = \text{Revenue} - \text{Variable Costs} = \$300,000 - \$30,000 = \underline{\$270,000}$$

Note: The bonus is a variable cost because it varies in proportion to the level of sales activity (i.e. number of hours of services provided to clients by Bean & Co). All other costs are fixed.

3. During the first six months in 20X2, Digitech had the following results:

	\$
Sales	809,000
Variable Costs	40,000
Fixed Costs	410,000
Operating profit	359,000

Compute the company's margin of safety percentage for the first six months in 20X2, round your answer to the nearest percentage.

Answer: [_____]

ANSWER

$$1 - \frac{\$410,000}{\$809,000 - \$40,000} = 47\% \text{ OR } \frac{\$359,000}{\$809,000 - \$40,000} = 47\% \text{ OR } \frac{\$809,000 - \text{BE Sales Dollars}}{\$809,000}$$

$$\text{Note: BE Sales Dollars} = \frac{\text{Fixed Costs}}{\text{CM Ratio}} = \frac{\$410,000}{\frac{(\$809,000 - \$40,000)}{\$809,000}}$$

$$\text{Or MoS\%} = 1/\text{DOL} = \text{Operating profit}/\text{CM} = 359,000/(\$809,000 - 40,000) = 47\%$$

4. Ken and Company expects the following operating results for next year:

Sales	\$400,000
Margin of safety	\$100,000
Contribution margin ratio	75%

Compute the expected Degree of Operating Leverage for next year.

Answer: [_____]

MUST SHOW WORKINGS

$$\$400,000 - \text{Breakeven sales} = \$100,000$$

$$\text{Breakeven sales} = \$300,000$$

$$\text{Breakeven sales} = \text{Fixed expenses} \div \text{Contribution margin ratio}$$

$$\$300,000 = \text{Fixed expenses} \div 0.75$$

$$\text{Fixed expenses} = \$225,000$$

$$\text{Contribution margin} = \$400,000 * 75\% = \$300,000$$

$$\text{Operating profit } \$300,000 - \$225,000 = \$75,000$$

$$\text{Degree of Operating Leverage } \$300,000/\$75,000 = 4$$

5. Chee and Company specializes in manufacturing a high end evening handbags with real crystals and other high quality materials. The total variable manufacturing cost per bag is \$50. Total monthly fixed manufacturing cost is \$6,000. The company uses an average cost per unit system, that is, all fixed manufacturing costs are spread among the actual total number of bags produced in the month. For the coming month, it is expected that the company will sell 1,000 bags and each at \$150. Assuming there is no beginning inventory, using absorption costing, what is the difference in gross profit for the company in the following two scenarios?

- A. Number of bags sold is 1,000 and number of bags made is also 1,000.
 B. Number of bags sold is 1,000 but number of bags made is higher at 1,200.

Answer: [Scenario B's gross profit is higher or lower (please delete one) by

[_____]

MUST SHOW WORKINGS

Production units	1,000	1,200	
Total fixed manufacturing costs	\$6,000	\$6,000	
Total variable manufacturing costs	\$50,000	\$60,000	
Total manufacturing costs	\$56,000	\$66,000	
Average manufacturing costs per unit	\$56	\$55	

Revenues		\$150,000		\$150,000
Beg inv	\$0		\$0	
Total variable manufacturing costs	\$50,000		\$60,000	
Total fixed manufacturing costs	\$6,000		\$6,000	
End inv	\$0		200*\$55=\$11,000	
Cost of goods sold		\$56,000		\$55,000
Gross Profit		\$94,000		\$95,000

Hence, gross profit is \$1,000 higher

6. CKY Consultancy applies overhead on the basis of professional hours. Given the following data, compute the under- or over-application of overhead for the period:

Estimated annual overhead cost	\$1,200,000
Actual annual overhead cost	\$1,145,000
Estimated professional hours	300,000
Actual professional hours	280,000

Answer: [_____]
 Under-applied/Over applied
 (Circle your answer)

ANSWER

Applied $\$1,200\text{k}/300\text{k} \times 280\text{k} = \$1,120\text{k}$
 Under-applied $= \$1,145\text{k} - \$1,120 = 25\text{k}$

7. SAG Partnership is a legal firm. The firm charges clients using a predetermined rate based on a monthly maximum capacity of 100,000 professional hours. In August, the expected and actual total overhead was \$1,000,000. Actual professional hours charged in August were 70,000 hours. Compute the amount of overhead to be treated as excess or idle capacity.

Answer: [_____]

ANSWER

Rate $= \$1,000,000/100,000 = \10 per professional hour
 Applied $= \$10 \times 70,000 = \$700,000$
 Excess capacity $= \$1,000,000 - \$700,000 = \$300,000$

Use the following information for questions 8-9:

TECH Singapore Limited is the sole distributor of a special coffee making machine imported from Europe. The purchase price of the machine from Europe has been quite stable at \$150 per unit.

In 20X1, the total annual fixed cost was \$900,000. In addition, TECH had to pay for the freight cost from Europe which was \$10 per unit. The selling price per unit was \$288 and 9,000 units of the coffee machine were sold.

8. For 20X2, the Sales Director feels that the company's operating profit can be improved if the company incurs an annual TV advertisement of \$200,000. The Sales Director proposes to increase the selling price slightly to \$308 to cover some of the increased advertising costs. If the Sales Director is confident that with the new TV advertisement, sales will go up from 9,000 units to more than 10,000 units in 20X2 even with the slightly higher price, should the company go ahead with the \$200,000 advertisement? Compute the change in profit with and without the proposed change by the Sales Director for 20X2 assuming that 10,000 units can be sold with the TV advertisement.

Answer: [_____]

MUST SHOW WORKINGS

	20X1	20X2
Units sold	9,000	10,000
Selling price	\$288	\$308
Cost per unit excluding freight	\$150	\$150
Freight cost per unit	\$10	\$10
Revenue	\$2,592,000	\$3,080,000
Cost of goods excluding freight	\$1,350,000	\$1,500,000
Freight cost	\$90,000	\$100,000
Contribution margin	\$1,152,000	\$1,480,000
Fixed cost	\$900,000	\$1,100,000
Operating income	\$252,000	\$380,000

\$128,000

9. What would have been the operating income of TECH for 20X1 if the units sold were 8,000 instead and the freight cost was 40% higher?

Answer: [_____]

MUST SHOW WORKINGS

Units sold	8,000
Selling price	\$288
Cost per unit excluding freight	\$150
Freight cost per unit	\$14
Revenue	\$2,304,000
Cost of goods excluding freight	\$1,200,000
Freight cost	\$112,000
Contribution margin	\$992,000
Fixed cost	\$900,000
Operating income	\$92,000

10. Rich Company has two service departments, Custodial Services and Maintenance department as well as two operating departments, Machinery and Milling. The CFO believes that variable and fixed service department costs should be charged to operating departments separately in order to provide more useful data for cost control. The maintenance department budgeted monthly fixed cost is \$30,000 and budgeted variable cost per square metre is \$0.5. The fixed costs of the Maintenance Department are determined based on peak-period demand. Data regarding the two operating departments are:

	Percentage of Peak Period Capacity Required	Actual Space occupied by each Department in January	Budgeted Space for each Department in January
Machining	30%	45,000	42,000
Milling	70%	90,000	98,000

How much Maintenance Department cost should be charged to Milling Department for performance evaluation purposes?

Answer: [_____]

$$\$0.5 \times 90,000 + 70\% \times \$30,000 = \$66,000$$

$$\$0.5 \times 90,000 + 98K / (42K + 98K) \times \$30,000 = \$66,000$$

For FC, can use peak period % or budgeted space, they give the same answer